

## BUILT\_IN FUNCTIONS

- 1) Write a query using the Production.Product table displaying the ProductID, Color, and Name columns. If the Color column contains a NULL value, replace the color with 'No Color'.

USE AdventureWorks\_Practice;

GO

SELECT

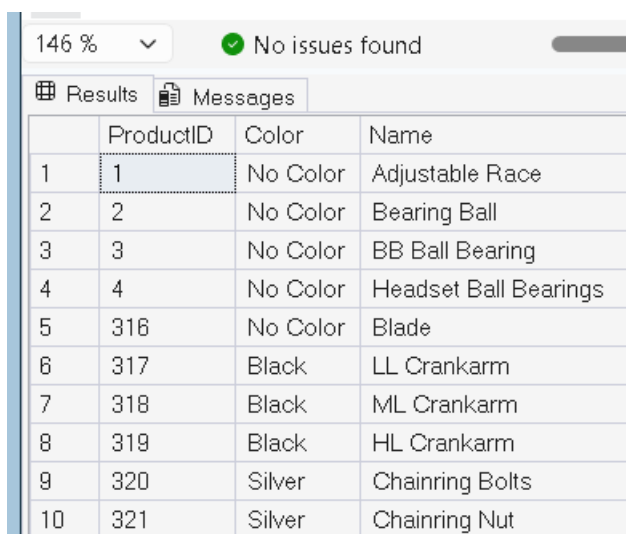
ProductID,

ISNULL(Color, 'No Color') AS Color,

Name

FROM Production.Product;

### Output



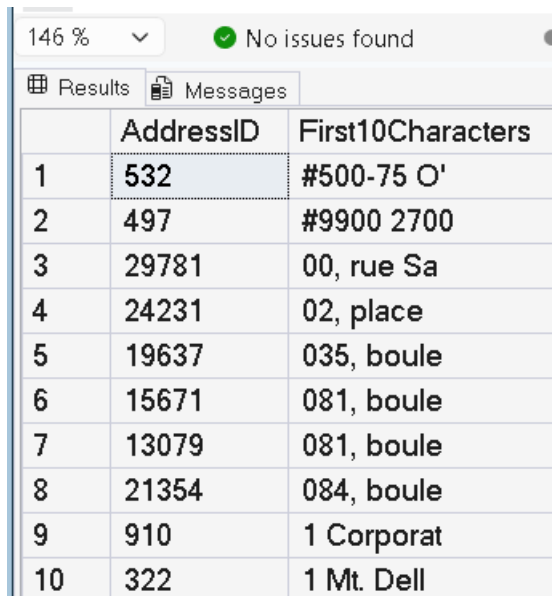
146 %    No issues found

|    | ProductID | Color    | Name                  |
|----|-----------|----------|-----------------------|
| 1  | 1         | No Color | Adjustable Race       |
| 2  | 2         | No Color | Bearing Ball          |
| 3  | 3         | No Color | BB Ball Bearing       |
| 4  | 4         | No Color | Headset Ball Bearings |
| 5  | 316       | No Color | Blade                 |
| 6  | 317       | Black    | LL Crankarm           |
| 7  | 318       | Black    | ML Crankarm           |
| 8  | 319       | Black    | HL Crankarm           |
| 9  | 320       | Silver   | Chainring Bolts       |
| 10 | 321       | Silver   | Chainring Nut         |

**2) Write a query that displays the first 10 characters of the AddressLine1 column in the Person.Address table.**

```
SELECT  
  
    AddressID,  
  
    LEFT(AddressLine1,10) AS First10Characters  
  
FROM Person.Address;
```

Output



The screenshot shows a SQL Server query results window. At the top, there is a zoom level of 146% and a status bar indicating 'No issues found'. Below the status bar, there are tabs for 'Results' and 'Messages'. The 'Results' tab is active, displaying a table with three columns: 'AddressID', 'First10Characters', and an unnamed column. The table contains 10 rows of data, numbered 1 through 10 in the first column. The 'AddressID' column contains values: 532, 497, 29781, 24231, 19637, 15671, 13079, 21354, 910, and 322. The 'First10Characters' column contains values: '#500-75 O'', '#9900 2700', '00, rue Sa', '02, place', '035, boule', '081, boule', '081, boule', '084, boule', '1 Corporat', and '1 Mt. Dell'.

|    | AddressID | First10Characters |
|----|-----------|-------------------|
| 1  | 532       | #500-75 O'        |
| 2  | 497       | #9900 2700        |
| 3  | 29781     | 00, rue Sa        |
| 4  | 24231     | 02, place         |
| 5  | 19637     | 035, boule        |
| 6  | 15671     | 081, boule        |
| 7  | 13079     | 081, boule        |
| 8  | 21354     | 084, boule        |
| 9  | 910       | 1 Corporat        |
| 10 | 322       | 1 Mt. Dell        |

**3) Write a query using the Sales.SpecialOffer table that multiplies the MaxQty column by the DiscountPct column. If MaxQty is NULL, replace it with the value 10. Include the SpecialOfferID and Description columns.**

```
SELECT  
  
    SpecialOfferID,
```

```

Description,

ISNULL(MaxQty,10) AS MaxQty,

DiscountPct,

ISNULL(MaxQty,10) * DiscountPct AS Result

FROM Sales.SpecialOffer;

```

## Output

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 Results

 Messages

|    | SpecialOffer... | Description                 | MaxQty | Discount... | Result |
|----|-----------------|-----------------------------|--------|-------------|--------|
| 1  | 1               | No Discount                 | 10     | 0.00        | 0.00   |
| 2  | 2               | Volume Discount 11 to 14    | 14     | 0.02        | 0.28   |
| 3  | 3               | Volume Discount 15 to 24    | 24     | 0.05        | 1.20   |
| 4  | 4               | Volume Discount 25 to 40    | 40     | 0.10        | 4.00   |
| 5  | 5               | Volume Discount 41 to 60    | 60     | 0.15        | 9.00   |
| 6  | 6               | Volume Discount over 60     | 10     | 0.20        | 2.00   |
| 7  | 7               | Mountain-100 Clearance S... | 10     | 0.35        | 3.50   |
| 8  | 8               | Sport Helmet Discount-2002  | 10     | 0.10        | 1.00   |
| 9  | 9               | Road-650 Overstock          | 10     | 0.30        | 3.00   |
| 10 | 10              | Mountain Tire Sale          | 10     | 0.50        | 5.00   |

**4) Write a query that calculates the number of days between the OrderDate and ShipDate using the Sales.SalesOrderHeader table. Include the SalesOrderID, OrderDate, and ShipDate columns.**

```

SELECT

    SalesOrderID,

    OrderDate,

```

```

ShipDate,

DATEDIFF(DAY, OrderDate, ShipDate) AS DaysTaken

FROM Sales.SalesOrderHeader;

```

## Output

| 146 %            | ✓ No issues found | Ln: 6, Ch: 1            | SPC                     | CRLF      | L |
|------------------|-------------------|-------------------------|-------------------------|-----------|---|
| Results Messages |                   |                         |                         |           |   |
|                  | SalesOrderID      | OrderDate               | ShipDate                | DaysTaken |   |
| 1                | 43659             | 2011-05-31 00:00:00.000 | 2011-06-07 00:00:00.000 | 7         |   |
| 2                | 43660             | 2011-05-31 00:00:00.000 | 2011-06-07 00:00:00.000 | 7         |   |
| 3                | 43661             | 2011-05-31 00:00:00.000 | 2011-06-07 00:00:00.000 | 7         |   |
| 4                | 43662             | 2011-05-31 00:00:00.000 | 2011-06-07 00:00:00.000 | 7         |   |
| 5                | 43663             | 2011-05-31 00:00:00.000 | 2011-06-07 00:00:00.000 | 7         |   |
| 6                | 43664             | 2011-05-31 00:00:00.000 | 2011-06-07 00:00:00.000 | 7         |   |
| 7                | 43665             | 2011-05-31 00:00:00.000 | 2011-06-07 00:00:00.000 | 7         |   |
| 8                | 43666             | 2011-05-31 00:00:00.000 | 2011-06-07 00:00:00.000 | 7         |   |
| 9                | 43667             | 2011-05-31 00:00:00.000 | 2011-06-07 00:00:00.000 | 7         |   |
| 10               | 43668             | 2011-05-31 00:00:00.000 | 2011-06-07 00:00:00.000 | 7         |   |

**5) Write a query that displays the year and numeric month of each OrderDate. Include the SalesOrderID and OrderDate columns.**

```

SELECT

SalesOrderID,

OrderDate,

YEAR(OrderDate) AS OrderYear,

MONTH(OrderDate) AS OrderMonth

FROM Sales.SalesOrderHeader;

```

## Output

146 %   No issues found  Ln: 6, Ch: 29

 Results

 Messages

|    | SalesOrderID | OrderDate               | OrderYear | OrderMonth |
|----|--------------|-------------------------|-----------|------------|
| 1  | 43659        | 2011-05-31 00:00:00.000 | 2011      | 5          |
| 2  | 43660        | 2011-05-31 00:00:00.000 | 2011      | 5          |
| 3  | 43661        | 2011-05-31 00:00:00.000 | 2011      | 5          |
| 4  | 43662        | 2011-05-31 00:00:00.000 | 2011      | 5          |
| 5  | 43663        | 2011-05-31 00:00:00.000 | 2011      | 5          |
| 6  | 43664        | 2011-05-31 00:00:00.000 | 2011      | 5          |
| 7  | 43665        | 2011-05-31 00:00:00.000 | 2011      | 5          |
| 8  | 43666        | 2011-05-31 00:00:00.000 | 2011      | 5          |
| 9  | 43667        | 2011-05-31 00:00:00.000 | 2011      | 5          |
| 10 | 43668        | 2011-05-31 00:00:00.000 | 2011      | 5          |

**6) Write a query using the Sales.SalesOrderHeader table that displays the SubTotal rounded to 2 decimal places. Include the SalesOrderID.**

SELECT

SalesOrderID,

ROUND(SubTotal,2) AS RoundedSubTotal

FROM Sales.SalesOrderHeader;

Output

146 % ✓ No issues found

Results Messages

|    | SalesOrderID | RoundedSubTotal |
|----|--------------|-----------------|
| 1  | 43659        | 20565.62        |
| 2  | 43660        | 1294.25         |
| 3  | 43661        | 32726.48        |
| 4  | 43662        | 28832.53        |
| 5  | 43663        | 419.46          |
| 6  | 43664        | 24432.61        |
| 7  | 43665        | 14352.77        |
| 8  | 43666        | 5056.49         |
| 9  | 43667        | 6107.08         |
| 10 | 43668        | 35944.16        |

**7) Write a statement that generates a random number between 1 and 10 each time it is run.**

`SELECT FLOOR(RAND() * 10) + 1 AS RandomNumber;`

146 % ✓ No issues found

Results Messages

|   | RandomNumber |
|---|--------------|
| 1 | 6            |